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Reinforcement Learning-Based Virtual Time Compressed Mirror for Underwater Acoustical Channel Equalization Based on Main Path to Side Path Ratio Criterion

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Abstract—Underwater acoustical (UWA) channels exhibit severe multipath propagation and long delay spreads, resulting in serious intersymbol interference in communications. In this article, we propose a virtual time compressed mirror (VTCM), a sparse adaptive equalizer, and its reinforcement learning (RL)-based version, RLVTM. VTCM can compress the time-spread channel with a virtual mirror based on sparse estimation, while RLVTM enhances this approach by exploiting RL to adjust VTCM's parameter-setting according to the UWA environment adaptively. In addition, we introduce a main path to side path ratio (MSR) criterion to evaluate the equalization performance in multipath channels before demodulation. Simulations and experiments demonstrate that both VTCM and RLVTM significantly improve communication performance. MSR consistently reflects symbol error rate performance.

Index Terms—Channel equalization, main path to side path ratio (MSR), reinforcement learning (RL), underwater acoustical (UWA) communications, virtual time compressed mirror (VTCM).

I. INTRODUCTION

THE underwater acoustical (UWA) channel is characterized as a time-dispersive rapidly fading channel, suffering from extended multipath propagation. The long channel delay spread leads to significant intersymbol interference (ISI) in UWA communications [1], [2]. Channel equalization, which is to restore the transmitted information by combating severe ISI and distortion [3], [4], is very critical in UWA communications.

Conventional equalization is usually designed in the error sense, such as mean-squared error (mse). The mse criterion focuses on

minimizing the average squared error between the transmitted and equalized received signals, and is usually monotonic with the signal-to-noise ratio (SNR) criterion. To minimize the error more efficiently, recursive algorithms, such as least mean square (LMS) [5], recursive least square (RLS) [5], and normalized LMS (NLMS) [6], are widely used in practical systems.

Due to the long delay spread in UWA channels, equalizers often require many taps, leading to high computational complexity and large pilot overheads. To mitigate these issues, exploiting the natural sparsity of UWA channels becomes a promising approach [7]. Algorithms, such as the improved proportionate normalized least mean squares (IPNLMS) [8] and the improved proportionate affine projection algorithm (IPAPA) [9], have shown superior performance by leveraging sparsity [10], since the sparse channels usually result in wide-sense sparse equalizers [11], [12]. Moreover, sparse estimations are combined into UWA channel equalizers, such as the sparse adaptive orthogonal matching pursuit-APA [13] and the sparse adaptive subspace pursuit-IPAPA [14].

However, in practical UWA communications, it is the system's symbol error rate (SER), not the mse, that really matters. Various simulations have illustrated that minimizing the mse does not necessarily produce the minimum SER (MSER) performance. Therefore, MSER equalizers are proposed: MSER equalizer by brute-force search [15], and adaptive MSER (AMSER) equalizer [16], [17], [18].

Similarly to conventional equalization, considering the sparsity of the equalizers, the authors in [19] and [20] propose the sparsity-aware normalized AMSER equalizer by utilizing the IPNLMS algorithm. Further improvements are made by replacing NLMS with RLS algorithms [21]. The variable adaptive subgradient projection is employed to derive a sparse control MSER equalizer, showing that sparse methods outperform nonsparse MSER equalizers [22].

However, both nonsparse and sparse MSER equalizers involve complex procedures, such as modulation and demodulation; thus, it is difficult to obtain the explicit expressions of SER or bit error rate (BER) [21]. Moreover, there is generally no closed-form solution in the MSER sense [23]. Furthermore, even the suboptimal solution is usually with very high computational complexity.

An alternative to channel equalization in the error sense is the time reversal mirror (TRM) in the temporal compression of the channel sense. TRM can adaptively match the acoustical channel without any prior knowledge and make the signal energy focused at the receiver [24], [25], [26], [27], [28]. The virtual time reversal mirror (VTRM) compresses the multipath energy into a single path [29], [30], [31], [32] by simulating the time reversal process. It can avoid the SER or BER expression and mitigate the ISI by compressing the multiple paths in the UWA channels, thus only requires the first several steps of the receiver processing in UWA communications. However, the VTRM

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is highly dependent on the specific multipath structure owing to the channel reversal procedure.

Meanwhile, above equalizers, whether designed in the error sense or the temporal compression of the channel sense, require the setting of many parameters. Usually, the parameters should be adjusted as the UWA channels change. However, the channels vary both temporally and spatially, it is often very challenging to keep up with these changes [33].

With the development of artificial intelligence, researchers have increasingly turned to machine learning (ML) methods to address the problems in the complex environments. Reinforcement learning (RL), including deep RL, has emerged as a significant technique. In RL, an agent learns by trial and error to maximize cumulative rewards, enabling it to develop satisfactory policies without prior knowledge [34], [35], [36], [37]. This makes RL effective for decision-making problems in dynamic settings and inspires researchers to use it in complex UWA environments.

RL has been proven to be effective in various UWA communication problems. In adaptive transmission design, RL is employed to recursively estimate channel parameters, track channel dynamics, and optimize transmissions, ultimately minimizing long-term costs [38], [39]. It helps determine the suitable settings for transmission power, modulation order, and coding rate, enabling a tradeoff between energy consumption and transmission latency [38]. RL is also used to identify the tradeoff between throughput and power consumption [39].

For channel estimation, tracking, and equalization in the physical layer research, RL is efficient in adjusting the UWA channel parameters to improve the performance. In [40] and [41], RL-based channel tracking methods determine the time-varying rank of the subspace channel, enabling effective channel tracking for time-varying UWA channels. Wang et al. [42] used RL to predict the coordinates of nonzero elements in the sparse channel, thereby predicting and tracking the time-varying channel in the delay-Doppler domain. Guo et al. [43] proposed an RL-based channel estimation to learn channel sparsity and Doppler scale factor, which are crucial for UWA channels. He and Tao [44] dynamically selected the optimal number of taps and step size based on channel impulse response (CIR) and SNR conditions.

Given the aforementioned progress, we focus on the equalization of the UWA channels in this work and provide the contributions as follows.

- 1) We propose a virtual time compressed mirror (VTCM) approach in the temporal compression sense. Instead of temporal reversal of the UWA channel like VTRM, we use sparse estimation to identify the virtual mirror due to the inherited sparsity of the equalizers, thereby realizing the temporal compression efficiently. The VTCM's performance does not depend on the multipath structure like the VTRM; thus, VTCM is more robust in different UWA channels. Both channel estimation-employed VTCM and pilot-employed VTCM are studied. The pilot-employed VTCM avoids the channel estimation and realizes the channel equalization within one step.
- 2) We present the main path to side path ratio (MSR) criterion in the temporal compression sense. The MSR criterion, which uses the power delay profile (PDP) [45] of the multipath channels, can evaluate the multipath compression effect, and thereby can be used as a criterion of the equalization in multipath channels. MSR pays attention to the ISI with the help of the PDP, and thus is more consistent with SER/BER than mse, which is in an equally sampled error sense. It does not involve modulation and demodulation; therefore, it is easier to provide the explicit expressions of the MSRs under different equalizers than the SER/BER criterion.
- 3) We improve the proposed VTCM through RL. We propose an RL-based VTCM (RLVTCM) to adapt to the various channel conditions by introducing RL into the proposed VTCM. We take

the MSR as the RL reward. In RLVTCM, RL interacts with the UWA environment to maximize MSR, dynamically adjusting the VTCM's parameters to the complex UWA environment.

- 4) In addition, our simulations and experiments, conducted under both sparse and wide-sense sparse channels, demonstrate that VTCM and RLVTCM outperform traditional equalization algorithms, such as VTRM, MMSE, LMS, IPNLMS, RLS, and proportional RLS (PRLS) [46]. It is also proved that the proposed MSR criterion is consistent with the SER results in both simulations and experiments. Notably, the pilot-employed RLVTCM improves the VTCM through RL significantly, and achieves the highest MSR and the lowest SER among all the algorithms, owing to its interaction with UWA channels to obtain suitable parameters setting in the channel equalization and its avoidance of channel estimation steps.

By improving channel equalization in UWA communications, our work expands global ocean observing systems, helping to collect data more reliably and efficiently from complex underwater environments with UWA communications.

The rest of this article is organized as follows. Section II introduces the system model and multipath UWA channels and explains the basic principles of VTRM. Section III proposes the VTCM method and the MSR criterion. Section IV first reviews the RL and then proposes the RLVTCM. Computational complexity among different algorithms is given in Section V. Simulation and experimental results are given in Section VI and VII, respectively. Finally, Section VIII concludes this article.

II. SYSTEM MODEL AND CONVENTIONAL VTRM

The relationship of the transmitted signals $x(t)$ and the received signals $y(t)$ during the UWA communication is expressed as

$$y(t) = x(t) \otimes h(t) + n(t) \quad (1)$$

in which $\otimes$ denotes the convolution operation, $h(t)$ is the CIR, and $n(t)$ represents the ambient noise which is usually assumed as white Gaussian noise. The UWA channel is commonly multipath and time-variant. We assume that it has $N_{pa}(t)$ multipath arrivals, and each arrival is characterized by the amplitude $A_p(t)$ and time delay spread $\tau_p(t)$. Then, the CIR can be expressed as

$$h(t) = \sum_{p=1}^{N_{pa}(t)} A_p(t) \delta(t - \tau_p(t)). \quad (2)$$

In VTRM, the CIR in (2) is simplified as a time-invariant channel with N_{pa} multipath arrivals. Its amplitude is A_p , and the time delay spread is τ_p for the p th path. Then, the CIR in VTRM is simplified as

$$h_c(t) = \sum_{p=1}^{N_{pa}} A_p \delta(t - \tau_p). \quad (3)$$

Although the CIR is time-varying as in (2), in practice, we set a reasonable short frame or block length, then within each frame or block, the channel can be assumed time-invariant in practical UWA communications. Then, if we reverse the CIR on time as $h_c(-t)$, which represents the VTRM, then the convolution of this time-reversed channel with the CIR can be considered as a virtual channel after VTRM. We use a Q -function to describe this virtual channel as follows:

$$\begin{aligned} Q(t) &= h_c(t) \otimes h_c(-t) \\ &= \left[\sum_{p=1}^{N_{pa}} A_p \delta(t - \tau_p) \right] \otimes \left[\sum_{p=1}^{N_{pa}} A_p \delta(t + \tau_p) \right] \end{aligned}$$

$$\begin{aligned}
&= \sum_{p=1}^{N_{pa}} A_p^2 \delta(t) + \sum_{i=1}^{N_{pa}} \sum_{j=1, j \neq i}^{N_{pa}} A_i A_j \delta(t + \tau_i - \tau_j) \\
&\approx A \delta(t)
\end{aligned} \quad (4)$$

where the last approximation is made under the assumption that $\sum_{p=1}^{N_{pa}} A_p^2 \gg A_i A_j$, $j \neq i$, and A is considered as the equivalent amplitude of the virtual channel, and $A = \sum_{p=1}^{N_{pa}} A_p^2$. For conventional VTRM, the Q -function is assumed as an ideal single-path channel, which is approximated to the Dirac delta function.

In the practice of VTRM, the estimated channel $\hat{h}(t)$ is obtained from the pilots. The VTRM $\hat{h}(-t)$ is then applied on $y(t)$ to recover the transmitted signal as follows:

$$\begin{aligned}
\hat{x}(t) &= y(t) \otimes \hat{h}(-t) \\
&= [x(t) \otimes h(t) + n(t)] \otimes \hat{h}(-t) \\
&\approx x(t) \otimes Q(t) + \tilde{n}(t) \\
&\approx Ax(t)
\end{aligned} \quad (5)$$

where $\tilde{n}(t)$ is the noise after VTRM. The last approximation is made under the assumption that the SNR is high enough, allowing us to omit the contribution of $\tilde{n}(t)$.

Through (5), we can observe that VTRM only needs the first several steps of the receiver processing in UWA communications, and it can mitigate the ISI caused by multipath effect through the temporal compression of the channel without the SER or BER expression.

However, VTRM often fails in practical underwater applications due to the following reasons: VTRM is subject to the multipath shape of the channel. Different shapes of channels can lead to different Q -function of VTRM, sometimes it is far from a Dirac delta function, if the subpaths are comparably high to the main path, for example. In most practical systems, arrays are often necessary to strengthen the main path to enhance the VTRM's reliability, as VTRM with a single hydrophone usually does not work well. The performance of the VTRM is limited by the performance of channel estimation. The poor channel estimation leads to poor VTRM equalization. Thus, VTRM, which relies on simply reversing the estimated channel in the time-delay domain, is not efficient for UWA channel equalization.

III. PROPOSED VTCM

Inspired by the VTRM in the temporal compression of the channel sense, we propose a VTCM in this section. Instead of a time-reversed estimated channel, we design an equalization filter as VTCM $w(t)$ for channel compression. By exploiting the inherent sparsity in VTCM, we introduce two schemes to implement VTCM in multipath UWA channels.

A. Algorithm Design of VTCM

Scheme 1: Estimated-channel-employed VTCM

To compress the UWA multipath channels temporally, the proposed VTCM $w(t)$ should satisfy $h(t) \otimes w(t) = \delta(t)$. However, $h(t)$ is unknown at the receiver, so we employ channel estimation $\hat{h}(t)$ instead. We consider the channel within a reasonable short frame or block of data transmission duration, then the channel can be assumed as time-invariant as in (3). Then, after we rewrite the convolution into the sum of products after sampling, the VTCM should satisfy

$$\sum_{m=0}^{N_c-1} \hat{h}(k-m)w(m) = d(k), \quad k = 0, 1, \dots, N_d-1 \quad (6)$$

where $\hat{h}(k)$ represents the estimated CIR after sampling with $0 \leq k \leq N_c-1$, N_c represents the tap length of $\hat{h}(k)$, and $w(k)$ represents the VTCM after sampling and has the same length with $\hat{h}(k)$, resulting $N_d = 2N_c - 1$. $d(k)$ is a time-delayed and sampled Dirac delta function $\delta(t)$, in which $d(k) = 1$ if $k = N_c$, and $d(k) = 0$ otherwise.

Then, (6) in the matrix form is expressed as

$$\hat{\mathbf{H}}\mathbf{w} = \mathbf{d} \quad (7)$$

where $\mathbf{d} = [d(0), d(1), \dots, d(N_d-1)]^H \in \mathbb{C}^{N_d \times 1}$, $\mathbf{w} = [w(0), w(1), \dots, w(N_c-1)]^H \in \mathbb{C}^{N_c \times 1}$, and $\hat{\mathbf{H}} \in \mathbb{C}^{N_d \times N_c}$ is the convolution matrix derived from the estimated CIR $\hat{h}(t)$. $\hat{\mathbf{H}}$, also known as a Toeplitz matrix, is defined as

$$\hat{\mathbf{H}} = \begin{bmatrix} \hat{h}(0) & 0 & \dots & 0 \\ \hat{h}(1) & \hat{h}(0) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \hat{h}(N_c-1) & \hat{h}(N_c-2) & \ddots & \hat{h}(0) \\ 0 & \hat{h}(N_c-1) & \ddots & \vdots \\ \vdots & \vdots & \ddots & \hat{h}(N_c-2) \\ 0 & 0 & \dots & \hat{h}(N_c-1) \end{bmatrix}. \quad (8)$$

The proposed VTCM $\mathbf{w}$ can be estimated based on (7). Although minimum mse (MMSE) or least squares methods in the mse sense can be used to estimate VTCM, we consider that the long VTCM inherently possesses wide-sense sparsity in this work. Then, the K -sparse estimation is exploited for the VTCM estimation in (7).

Usually, the greedy matching pursuit algorithm, which employs dictionaries with finer delay resolution (subsample spacing), addresses the sparsity estimation in (7). In this context, we employ the generalized orthogonal matching pursuit (GOMP) algorithm [47]. GOMP, an extension of the orthogonal matching pursuit (OMP), is the most prevalent orthogonal variant of the greedy matching pursuit used for reconstructing K -sparse signals. By selecting multiple "correct" indices, GOMP completes the K -sparse estimation with significantly fewer iterations compared to OMP, thereby reducing complexity. In addition, it has been proven to be effective for wide-sense sparse signals and in the presence of noisy measurements [48], therefore can work under either not very sparse channel conditions or low SNR conditions.

The procedure of estimating the VTCM $\mathbf{w}$ by GOMP is as follows.

Assume that VTCM $\mathbf{w}$ is K ($K \ll N_c$)-sparse, where $K = \|\mathbf{w}\|_0$. The index set of VTCM is $\mathcal{T} = \{i \mid i \in \Omega, w(i) \neq 0\}$, where $\Omega = \{0, 1, \dots, N_c-1\}$ and $|\mathcal{T}| = K$. VTCM algorithm aims to identify these K "correct" indices. In the n th iteration, the correlations between the columns of $\hat{\mathbf{H}}$ and the modified measurements (residual) are compared. The D indices of the columns corresponding to the D largest correlations are identified. Denoting the D indices as $\mathcal{J} = \{\phi(1), \dots, \phi(D)\}$, where

$$\phi(i) = \arg \max_{j: j \in \Omega, j \notin \{\phi(1), \dots, \phi(i-1)\}} |\langle \mathbf{r}_{n-1}, \hat{\mathbf{h}}_j \rangle| \quad (9)$$

and $\hat{\mathbf{h}}_j$ is the corresponding column of matrix $\hat{\mathbf{H}}$. Then, the number of the indices D is also the step size in each iteration. The extended support set as the n th iteration becomes $\Lambda_n = \Lambda_{n-1} \cup \mathcal{J}$. Through the support set, the solution is

$$\hat{\mathbf{w}}_{\Lambda_n} = \arg \min_{\mathbf{w}} \|\mathbf{d} - \hat{\mathbf{H}}_{\Lambda_n} \mathbf{w}\|_2. \quad (10)$$

Then, the residual $\mathbf{r}_n$ is updated. These operations are repeated until either the iteration number N_{it} reaches $\min(K, (N_d/D))$ or

the terminal residual falls below a threshold ($\|\mathbf{r}_{N_{it}}\|_2 \leq \xi$). Due to the orthogonality, indices in Λ_n cannot be reselected in succeeding iterations. The cardinality of Λ_s becomes DN_{it} eventually, where Λ_s is the final support set of $\hat{\mathbf{w}}$. Therefore, the output $\hat{\mathbf{w}}_{\Lambda_s}$ is obtained.

Scheme 2: Pilot-employed VTCM

In the previous scheme, we need to estimate the channel by pilots before the VTCM, therefore can obtain $\hat{\mathbf{H}}$. The channel estimation through pilots can introduce cumulative errors and compromise the performance. To address this issue, we propose a pilot-employed VTCM, and the equalization can be done within one step. We assume the channel time-invariant as in (3) within a frame or block length.

Assume the pilots

$$\mathbf{x}_p = [x_p(0), x_p(1), \dots, x_p(N_p - 1)]^H \in \mathbb{C}^{N_p \times 1}.$$

We define $\mathbf{x} = [\mathbf{0}_{(N_c-1) \times 1}^T, \mathbf{x}_p^H, \mathbf{0}_{(N_c-1) \times 1}^T]^H$. The K -sparse pilot-employed VTCM $\mathbf{w}$ should satisfy

$$\mathbf{x} = \mathbf{Y}\mathbf{w} \quad (11)$$

where $\mathbf{Y} \in \mathbb{C}^{(N_y+N_c-1) \times N_c}$ represents the convolution matrix from the received pilot $\mathbf{y}_p \in \mathbb{C}^{N_y \times 1}$, where $N_y = N_p + N_c - 1$. It is defined as follows:

$$\mathbf{Y} = \begin{bmatrix} y_p(0) & \overbrace{0 \quad \dots \quad 0}^{N_c-1} \\ y_p(1) & y_p(0) & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ \vdots & \vdots & \ddots & y_p(0) \\ y_p(N_y-1) & y_p(N_y-2) & \ddots & \vdots \\ 0 & y_p(N_y-1) & \ddots & \vdots \\ \vdots & \vdots & \ddots & y_p(N_y-2) \\ 0 & 0 & \dots & y_p(N_y-1) \end{bmatrix} \quad (12)$$

Given the wide-sense sparsity, GOMP is exploited to estimate the VTCM in (11) as in Scheme 1.

The pilot-employed VTCM algorithm with GOMP is shown in Algorithm 1. In Scheme 2, the D indices are as defined by

$$\phi(i) = \arg \max_{j: j \in \Omega, j \notin \{\phi(i-1), \dots, \phi(1)\}} |\langle \mathbf{r}_{n-1}, \mathbf{y}_j \rangle| \quad (13)$$

where $\mathbf{y}_j$ is the corresponding column of matrix $\mathbf{Y}$. The number of iterations $N_{it} = \min(K, (N_p + 2N_c - 2)/D)$. The corresponding VTCM is obtained from $\hat{\mathbf{w}}_{\Lambda_n} = \arg \min_{\mathbf{w}} \|\mathbf{x} - \mathbf{Y}_{\Lambda_n} \mathbf{w}\|_2$.

Compared with VTRM, the proposed VTCM algorithms do not rely on the assumption that $\sum_{p=1}^{N_{pa}} A_p^2 \gg A_i A_j$, $j \neq i$ in (4). This assumption is often difficult to satisfy in real-world UWA channels, as the main path is not always powerful enough. Consequently, VTCMs can offer superior performance over VTRM in many practical UWA channel scenarios.

Next, we compare the two VTCM schemes. Theoretically, in the absence of noise, $\mathbf{y} = \mathbf{H}\mathbf{x}$, (11) is equivalent to (7). It means that the performance of Scheme 1 would be equivalent to that of Scheme 2 if the perfect channel state information is available, without noise. However, due to the inherent noise in real-world conditions, $\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$, the two schemes are not identical. The VTCM $\mathbf{w}$ in Scheme 2 equalizes not only the channel but also the noise at the receiver, facilitating the recovery of the transmitted signals. As a result, Scheme 2 can offer improved performance.

Algorithm 1: Pilot-Employed VTCM Algorithm.

- 1: **Input:** The reconstruct matrix $\mathbf{Y}$, pilot vector $\mathbf{x}$, sparsity level K of VTCM $\mathbf{w}$ and step size or number of indices D ($D \leq K$) for each iteration;
- 2: **Initialize:** $\mathbf{r}_0 = \mathbf{x}$, $\Lambda_0 = \emptyset$, iteration count $n = 1$;
- 3: **while** $\|\mathbf{r}_n\|_2 > \xi$ and $n < \min\{K, \frac{(N_p + 2N_c - 2)}{D}\}$ **do**
- 4: Denote the D indices as $\mathcal{J} = \{\phi(1), \dots, \phi(D)\}$, where $\phi(i) = \arg \max_{j: j \in \Omega, j \notin \{\phi(i-1), \dots, \phi(1)\}} |\langle \mathbf{r}_{n-1}, \mathbf{y}_j \rangle|$;
- 5: Update the indices $\Lambda_n = \Lambda_{n-1} \cup \mathcal{J}$, and $\mathbf{Y}_{\Lambda_n}$;
- 6: Obtain solution $\hat{\mathbf{w}}_{\Lambda_n} = (\mathbf{Y}_{\Lambda_n}^H \mathbf{Y}_{\Lambda_n})^{-1} \mathbf{Y}_{\Lambda_n}^H \mathbf{x}$;
- 7: Update the residual $\mathbf{r}_n = \mathbf{x} - \mathbf{Y}_{\Lambda_n} \hat{\mathbf{w}}_{\Lambda_n}$;
- 8: $n = n + 1$;
- 9: **end while**
- 10: **Return:** the VTCM $\hat{\mathbf{w}}_{\Lambda_s}$, and equalized signal $\hat{\mathbf{x}} = \mathbf{Y} \hat{\mathbf{w}}_{\Lambda_s}$.

B. MSR Performance Analysis

To analyze the performance of the proposed equalization algorithms, considering the PDP, we define the MSR for multipath channels as follows:

$$\text{MSR} = \frac{P_q}{\sum_{p=1, p \neq q}^{\infty} |\tau_p - \tau_q| P_p} \quad (14)$$

where P_p represents the power of p th path, τ_p is the time delay spread of the p th path, τ_q is the time delay spread of the main path, and P_q is the power of the main path. In this work, we define the main path as the most powerful path. Notably, the most powerful path is neither necessarily the direct path [49], [50], nor is it always the first arrival path [50], [51], based on the unique characteristics of UWA channels. The denominator in (14) is the PDP of all the other paths, describing the distribution of power received over various time delays [45].

Since a higher MSR indicates less ISI effect on communications in the multipath channels, MSR can demonstrate the performance of the channel equalization. The MSR can be obtained right after equalization. It does not need further processing, such as demodulation and decoding, compared with SER/BER. Compared to MSE, MSR can omit the trivial details of the waveforms, and can evaluate the equalization well even in a low SNR.

Based on the definition in (14), the MSR of the time-varying multipath UWA channels, as described in (2), can be expressed as follows:

$$\text{MSR}_C(t) = \frac{A_q^2(t)}{\sum_{p=1, p \neq q}^{N_{pa}(t)} |\tau_p(t) - \tau_q(t)| A_p^2(t)}. \quad (15)$$

If we only consider the channel in a very short time, such as during one frame of data transmission, then we can assume the channel time-invariant as in (3). In this case, MSR_C can be simplified into (14).

In the practical perspective, it is difficult to obtain the path parameters, especially in a noisy UWA environment. To make the MSR more practical, we treat every tap in the CIR as a path. Then, the MSR becomes

$$\text{MSR}_C = \frac{h^2(q)}{\sum_{k=0, k \neq q}^{N_c-1} |k - q| T_s h^2(k)} \quad (16)$$

where the q th tap is the most powerful tap in CIR. Similarly, if the equalized channel is $\mathbf{h}_{eq} \in \mathbb{C}^{N_{eq} \times 1}$, then the MSR after equalization is as follows:

$$\text{MSR}_{eq} = \frac{h_{eq}^2(q)}{\sum_{k=0, k \neq q}^{N_{eq}-1} |k - q| T_s h_{eq}^2(k)} \quad (17)$$

in which $h_{\text{eq}}(q)$ is the amplitude of the most powerful tap and N_{eq} is the channel length in taps.

Based on the above definitions, we can evaluate different equalizers.

For the conventional VTRM, the Q -function (4) represents the equivalent multipath channel after the VTRM, and then the MSR according to the Q -function is as follows:

$$\text{MSR}_{\text{VTRM},Q} = \frac{\left(\sum_{p=1}^{N_{\text{pa}}} A_p^2\right)^2}{\sum_{i=1}^{N_{\text{pa}}} \sum_{j=1, j \neq i}^{N_{\text{pa}}} |\tau_i - \tau_j - \tau_{N_{\text{pa}}}| A_i^2 A_j^2}. \quad (18)$$

In VTRM, the amplitude at the main path tap after the equalization $h_{\text{VTRM}}(N_c - 1) = \sum_{k=0}^{N_c-1} h^2(k)$. The practical MSR in channel-tap form, based on (17), is then given by

$$\text{MSR}_{\text{VTRM}} = \frac{\left(\sum_{k=0}^{N_c-1} h^2(k)\right)^2}{\sum_{i=0}^{N_c-1} \sum_{j=0, j \neq i}^{N_c-1} |i - j - N_c + 1| T_s h^2(i) h^2(j)}. \quad (19)$$

From MSR_C in (16) and MSR_{VTRM} in (19), we can express MSR_{VTRM} as a function of the MSR_C as follows:

$$\text{MSR}_{\text{VTRM}} = \frac{\left(\sum_{k=0, k \neq q}^{N_c-1} (\text{MSR}_C |k - q| T_s + 1) h^2(k)\right)^2}{\sum_{i=0}^{N_c-1} \sum_{j=0, j \neq i}^{N_c-1} |i - j - N_c + 1| T_s h^2(i) h^2(j)}. \quad (20)$$

From (20), generally, the MSR_{VTRM} monotonically increases with MSR_C . This indicates that the performance of VTRM depends on the multipath UWA channels. According to the theory of VTRM, VTRM reconstructs multipath channels in a manner of matched filtering [52], which can maximize the output SNR at a given time instant through the filter, but is not designed for eliminating ISI. In fact, the matched filtering can increase the temporal dispersion sometimes, i.e., the duration of the overall impulse response of the system, so the ISI may still be serious after VTRM. In some channel conditions where the majority of energy resides in the main path and the temporal dispersion is negligible, VTRM may exhibit good performance. However, even in these conditions, if the SNR is low and temporal dispersion is long, VTRM may not work well, either. Therefore, the equalization performance of traditional VTRM heavily relies on the UWA channel condition, making it ineffective in severe UWA environments.

To observe the performance of VTCM, we first assume that the CIR is perfectly known in VTCM, which is the same as in (19) for the VTRM, then Schemes 1 and 2 are equivalent. With the estimated VTCM $\hat{\mathbf{w}} = [\hat{w}(0), \hat{w}(1), \dots, \hat{w}(N_c - 1)]^T$, the MSR for the VTCM can be expressed as follows:

$$\text{MSR}_{\text{VTCM}} = \frac{\left(\sum_{k=0}^{N_c-1} h(k) \hat{w}(N_c - 1 - k)\right)^2}{\sum_{i=0}^{N_c-1} \sum_{j=0, j \neq i}^{N_c-1} |i - j - N_c + 1| T_s h^2(i) \hat{w}^2(N_c - 1 - j)}. \quad (21)$$

According to (6), when the equivalent channel after VTCM is more similar to an ideal impulse response, the MSR is higher. A higher MSR results in better suppression of ISI, leading to a lower SER/BER.

To consider the practical VTCM in Scheme 1, since perfect CSI is not available, we have to employ the channel estimation to obtain $\hat{h}(i)$. Thus, the MSR for the VTCM in Scheme 1 can be expressed as follows:

$$\begin{aligned} \text{MSR}_{\text{VTCM_sch1}} &= \frac{\left(\sum_{k=0}^{N_c-1} \hat{h}(k) \hat{w}(N_c - 1 - k)\right)^2}{\sum_{i=0}^{N_c-1} \sum_{j=0, j \neq i}^{N_c-1} |i - j - N_c + 1| T_s \hat{h}^2(i) \hat{w}^2(N_c - 1 - j)}. \end{aligned} \quad (22)$$

Since Scheme 2 employs pilots to perform equalization within one step, and there is no channel estimation, we derive the equivalent equalized channel as follows: we use the chirp signal [linear frequency modulation (LFM) or hyperbolic frequency modulation], which is typically employed as the preamble signal in UWA communications, as $\mathbf{x}_p$. Based on its characteristics [53], the equivalent equalized channel can be expressed as follows:

$$\begin{aligned} h_{\text{eq_sch2}}(k) &= \sum_{m=0}^{N_p-1} \hat{x}(k - m) x_p(N_p - 1 - m), \\ k &= 0, 1, \dots, N_{\text{eq_sch2}} - 1 \end{aligned} \quad (23)$$

where $N_{\text{eq_sch2}} = 2N_c + 2N_p - 3$. According to (17), we have $q = N_c + N_p - 2$; therefore, the MSR in Scheme 2 is as follows: Under the condition that the main path in the equalized channel $\mathbf{h}_{\text{eq}}$ is significantly stronger than the side paths, the MSRs of Scheme 2 (24) shown at the bottom of this page and Scheme 1 (22) are approximately equivalent. In both schemes, a higher MSR is observed when the power is more concentrated in the main path and the delay spread is smaller. This indicates more effective mitigation of ISI, which corresponds to a lower SER.

IV. ENHANCED RLVTM

In this section, RL is employed to learn the UWA channel parameters during the VTCM. In the proposed VTCM algorithms, prior knowledge of sparsity K is crucial. However, the sparsity K may vary across different UWA channels and is difficult to know before each UWA communication. From the above algorithms, $K = D \times N_{\text{it}}$. It means that the pair of D , which is the step size in each iteration, and the number of iterations N_{it} should be carefully set according to the different characteristics of the UWA channels.

A typical RL algorithm includes the agent, states, actions, and reward function. In RL, an agent aims to maximize the received reward when interacting with complex, uncertain environments. At the n th time slot, the agent observes the current state $s_n \in \mathcal{S}$ of the environment, and then selects action $a_n \in \mathcal{A}$ according to the policy π . Then, the environment is updated to the next state s_{n+1} and the agent obtains the reward $r_{n+1} \in \mathcal{R}(s, a)$, where the reward serves as a function of the state $s \in \mathcal{S}$ and the agent's actions $a \in \mathcal{A}$. In general, the policy π is a mapping

$$\begin{aligned} \text{MSR}_{\text{VTCM_sch2}} &= \frac{h_{\text{eq_sch2}}^2(q)}{\sum_{k=0, k \neq q}^{N_{\text{eq_sch2}}-1} |k - q| T_s h_{\text{eq_sch2}}^2(k)} \\ &= \frac{\left(\sum_{k=0}^{N_p-1} \hat{x}(N_c + N_p - 2 - k) x_p(N_c - 1 - k)\right)^2}{\sum_{i=0, i \neq N_c + N_p - 2}^{2N_c + 2N_p - 4} |i - N_c - N_p + 2| T_s \left(\sum_{m=0}^{N_p-1} \hat{x}(i - j) x_p(N_c - 1 - j)\right)^2}. \end{aligned} \quad (24)$$

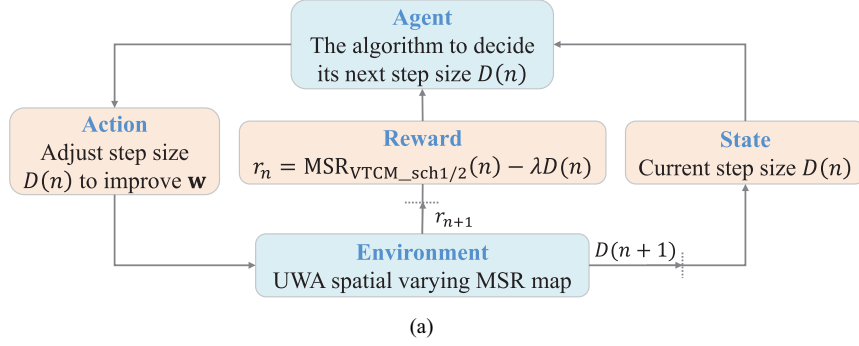


Fig. 1. Chart displaying the workflow of proposed RLVTM. The closed-loop agent (the algorithm to decide its next step size in RLVTM) interacts with the uncertain, potentially complex environment (the UWA spatially varying MSR map in RLVTM). The agent sends selected actions (the adjustment of step size) based on the current state (the current step size) to the environment, and then in turn receives the new state and the delayed reward (the equalization performance evaluated by MSR).

from states to a probability distribution over actions. When the agent following the policy π at time slot n , the policy π is $p(a_n = a | s_n = s)$. The goal of RL is to find an optimal policy π^* that achieves the maximum expected return from all states.

In this work, we use RL to find the optimal channel parameters that are most suitable to the uncertain and complex UWA communication environment and propose the RLVTM. We adapt D into $D(n)$, which varies in each iteration, to make the adaptation to the UWA channels more flexible. Through the RL, we perceive $D(n)$ and finally the sparsity K of the UWA channels. The agent, states, actions, and reward function in the RLVTM, as shown in Fig. 1, are defined as follows.

Agent: The agent is the ML algorithm (or the autonomous system) in RL. In the proposed RLVTM, the agent is the algorithm to decide its next step size $D(n+1)$. At the n th step, the agent observes current step size $D(n)$ (the current state s_n) and then selects action a_n to update the state into s_{n+1} according to the policy π . The UWA environment then provides feedback to the agent in the form of rewards r_{n+1} based on the state s_{n+1} .

Environment: The environment represents the adaptive problem space, including variables, boundary conditions, and valid actions. In the proposed RLVTM algorithm, the environment is represented by the UWA spatially varying MSR map, which the agent interacts with to optimize the VTCM through $D(n)$.

State: The state of the environment at a given time or iteration refers to the information that the agent perceives from the environment. The state space refers to all of the information provided by an environment's state. In the proposed RLVTM, we define the state space $\mathcal{S}$ as all the possible step size $D(n) \in \mathbb{N}^+$, where $\mathbb{N}^+$ denotes the positive integers. The state space $\mathcal{S}$ is

$$\mathcal{S} = \{1, 2, \dots, D_{\max}\} \quad (25)$$

where $D_{\max}$ is the maximum of $D(n)$. Then, the state $s_n = D(n) \in \mathcal{S}$. Note that the step size can vary at the different iteration n to make the algorithm more flexible to the UWA environment than VTCM without RL.

Action: The action is a step that the RL agent takes to navigate the environment. The action space denotes all possible actions the agent may take within a state in RL. In the proposed RLVTM, the action space $\mathcal{A}$ is defined as

$$\mathcal{A} = \{s - C, \dots, s - 1, s, s + 1, \dots, s + C\} \quad (26)$$

where $C \in \mathbb{N}^+$ is the maximum scale of the adjustments. Through the selected action $a_n \in \mathcal{A}$, we adjust $D(n)$ during the n th iteration.

Reward function: The reward is the feedback to the agent from the postaction environment based on the state. In the proposed RLVTM algorithm, the reward is the feedback from the UWA environment based on the state. It is usually related to the objective of the algorithm, as RL aims to maximize the reward. In this work, we set the objective function as the MSR of the channel after the VTCM in (24). To reduce the computational complexity, a regularization term is added to each iteration. Consequently, the corresponding reward function, based on the MSR in Scheme 2 (24) is formulated as

$$r_n = \text{MSR}_{\text{VTCM_sch2}}(n) - \lambda D(n) \quad (27)$$

where $\lambda D(n)$ provides the penalty of the computational complexity in the n th iteration.

RLVTM maximizes the MSR while allowing for a certain margin of error. The termination condition for each training episode is defined as follows: the iteration terminates if

$$\text{MSR}_{\text{VTCM}}(n) - \text{MSR}_{\text{VTCM}}(n-1) < \Delta$$

where $\Delta < 0$ represents the allowable margin of error. Then, the iteration terminates at N_{it} if this condition is not met. Then, the rewards are accumulated through iterations, resulting in the overall return

$$R = \sum_{n=1}^{N_{\text{it}}} \gamma^{n-1} r_n \quad (28)$$

in which $\gamma (0 \leq \gamma \leq 1)$ is the discount factor. In RLVTM, the sparsity K is expressed as follows:

$$K = \sum_{n=1}^{N_{\text{it}}} D(n). \quad (29)$$

RL aims to find an optimal policy that maximizes the expected return from all states. The state-action value or quality function $Q^\pi(s, a)$ is used to describe the goal

$$Q^\pi(s, a) = \mathbb{E}[R | s, \pi]. \quad (30)$$

The best policy, inferred from $Q^\pi(s, a)$, is determined by selecting actions a greedily at each state: $\arg \max_a Q^\pi(s, a)$. The corresponding Q-value $Q(s, a)$ represents the reward of a state-action pair. A comprehensive overview of RL is provided in [34].

In the context of RLVTM, we obtain optimal step size $D(n)$ to reconstruct the K -sparsity VTCM $\mathbf{w}$ with reasonable complexity. In RL, various methods can be employed, including nondeep learning methods, such as Q-learning and SARSA, or deep learning-based approaches, such as deep Q-learning. In this article, we adopt an ϵ -greedy

Algorithm 2: Pilot-Employed RLVTM Algorithm.

- 1: **Input:** The reconstruct matrix $\mathbf{Y}$, pilot vector $\mathbf{x}$;
- 2: **Output:** the VTCM $\hat{\mathbf{w}}$;
- 3: **Initialize:** the indices $\Lambda_0 = \emptyset$, states (25), actions (26), reward function (27) and Q-table $Q(s, a)$ (31), learning rate $\alpha \in (0, 1)$, discount factor $\gamma \in (0, 1)$, greedy probability $\epsilon \in (0, 1)$, decay factor $\eta \in (0, 1]$, maximum episodes I , and training episode $i = 1$;
- 4: **while** $i \leq I$ and QL is not converged **do**
- 5: Initial step size $D(1)$ and residual $\mathbf{r}_0 = \mathbf{x}$, iteration count $n = 1$, the cardinality of final support set $N_D = 0$;
- 6: **while** $N_D < M$ and $\text{MSR}_{\text{VTCM_sch2}}(n) - \text{MSR}_{\text{VTCM_sch2}}(n-1) \leq \Delta$ **do**
- 7: Select a_n via dynamic ϵ -greedy strategy;
- 8: Execute a_n and updating the index set $\mathcal{J}$;
- 9: Update the indices $\Lambda_n = \Lambda_{n-1} \cup \mathcal{J}$, and $\mathbf{Y}_{\Lambda_n}$;
- 10: Obtain solution $\hat{\mathbf{w}}_{\Lambda_n} = (\mathbf{Y}_{\Lambda_n}^H \mathbf{Y}_{\Lambda_n})^{-1} \mathbf{Y}_{\Lambda_n}^H \mathbf{x}$;
- 11: Update the residual $\mathbf{r}_n = \mathbf{x} - \mathbf{Y}_{\Lambda_n} \hat{\mathbf{w}}_{\Lambda_n}$;
- 12: Obtain next step size $D(n+1)$;
- 13: Calculate the $\text{MSR}_{\text{VTCM_sch2}}(n+1)$ (24) from $\mathbf{Y} \hat{\mathbf{w}}_{\Lambda_n}$ and reward r_{n+1} ;
- 14: Update Q-table (31);
- 15: $N_D = N_D + D(n)$;
- 16: $n = n + 1$;
- 17: **end while**
- 18: Update ϵ for next round $\epsilon = \eta\epsilon$;
- 19: $i = i + 1$;
- 20: **end while**
- 21: **Return:** the adaptive step size $D(n)$, VTCM $\hat{\mathbf{w}}_{\Lambda_s}$ with step 10, and equalized signal $\hat{\mathbf{x}} = \mathbf{Y} \hat{\mathbf{w}}_{\Lambda_s}$.

Q-learning scheme. In Q-learning, an agent chooses the optimal policy π based on the Q-values. For the purposed of $\arg \max_a Q^\pi(s, a)$, the action value function can be updated as follows:

$$Q(s_n, a_n) \leftarrow Q(s_n, a_n) + \alpha \left[r_{n+1} + \gamma \max_a Q(s_{n+1}, a) - Q(s_n, a_n) \right] \quad (31)$$

where $\alpha(0 \leq \alpha \leq 1)$ represents the learning rate of Q-learning, $Q(s_n, a_n)$ is Q-value of the selected action a_n under state s_n , and $Q(s_{n+1}, a)$ is the possible values after taking possible actions a at the $n+1$ th iteration. The RLVTM algorithm procedure is listed in Algorithm 2. To balance the exploration and exploitation of RLVTM, we use $\eta(0 < \eta \leq 1)$ as the decay factor of ϵ .

Note that it is theoretically possible to use RL to directly determine $\mathbf{w}$, but the action and state spaces would become very large in less sparse channels (with larger K) and with longer tap length (larger N_c), as commonly encountered in UWA channels with complex, long delay spreads. This results in a combinatorial explosion, making it impractical for RL to directly optimize $\mathbf{w}$. Even with deep RL, the training process would suffer from slow convergence and instability [35]. Therefore, in the proposed VTCM, we adopt this hybrid approach: RL is used to optimize $D(n)$, while sparse recovery algorithms, such as GOMP, are employed to optimize $\mathbf{w}$. This strategy balances adaptability and computational feasibility, avoiding the complexity associated with directly optimizing $\mathbf{w}$ through RL.

V. COMPUTATIONAL COMPLEXITY

We analyze the computational complexity of different algorithms in Table I. Computational complexity is compared in terms of floating-point operations (FLOPs).

TABLE I
COMPARISON OF COMPUTATIONAL COMPLEXITY

Methods	Computational complexity (FLOPs)
MMSE	$\frac{64}{3} N_c^3$
LMS	$(16K_f + 2)N_p$
IPNLMS	$(34K_f + 8)N_p$
RLS	$(26K_f^2 + 22K_f + 1)N_p$
PRLS	$(26K_f^2 + 31K_f + 8)N_p$
VTRM	$2N_c$
VTCM Scheme 1	$8N_{it}(2N_c - 1)N_c + N_{it}(8D^2 + 6D)2N_c - 1$
VTCM Scheme 2	$8N_{it}(N_p + 2N_c - 2)N_c + N_{it}(8D^2 + 6D)(N_p + 2N_c - 2)$
RLVTM Scheme 1	$8N_{it}(2N_c - 1)N_c + \sum_{n=1}^{N_{it}} n(16D(n)^2 + 12D(n))(2N_c - 1)$
RLVTM Scheme 2	$8N_{it}(N_p + 2N_c - 2)N_c + \sum_{n=1}^{N_{it}} n(16D(n)^2 + 12D(n))(N_p + 2N_c - 2)$

The computational complexity in both VTCM and RLVTM depends on the iteration number N_{it} , the channel length N_c , and the step size D in VTCM or $D(n)$ in RLVTM. In the proposed algorithm, each complex-valued addition contributes two FLOPs (real part and imaginary part), and each complex-valued multiplication requires six FLOPs (four real-valued multiplications and two real-valued additions). The complexity of inner products, matrix multiplications, and matrix inversions is scaled accordingly.

For LMS, IPNLMS, RLS, and PRLS, the iteration number of filters is equivalent to the length of pilot symbols N_p , and the tap length of filter K_f typically scales with the time delay spread. The computational complexity of LMS and IPNLMS are based on the work in [8] and [20], whereas RLS and PRLS are based on the work in [46] and [58].

Generally, the complexity of the proposed algorithms can exceed the complexity of VTRM, LMS, and IPNLMS, but may be lower than that of RLS, PRLS, and MMSE. Note that D and N_{it} must be predefined since the sparsity level K is often an empirical value in UWA communications in VTCM. A larger D leads to higher overhead in VTCM. Different from VTCM, the complexity of RLVTM varies with dynamical step size $D(n)$. Owing to the dynamic adjustments and the penalties for excessively large step sizes, RLVTM based on Q-learning may have lower complexity than VTCM after training.

In addition, in Scheme 2, pilot length N_p also needs to be considered. In Table I, Scheme 2 appears more complex than Scheme 1 due to the consideration of the pilot. However, Scheme 1 needs to work with the channel estimation. Taking MMSE channel estimation as an example, combining Scheme 1 with channel estimation, the complexity would be much higher than Scheme 2.

VI. SIMULATION RESULTS AND ANALYSIS

We conduct simulations under two typical channel conditions: the very sparse multipath channel and the wide-sense sparse (less sparse) multipath channel, as shown in Fig. 2. Both conditions are commonly encountered in UWA communications and can be simulated using channel models [54]. The very sparse multipath channel is characterized by most of the channel energy is concentrated in a few delays [7]. As a result, the CIR vectors contain more elements with values near or exactly zero than nonzero elements. This aligns with the mathematical concept of “sparsity.”

However, strict sparsity in UWA channels is not always guaranteed. These channels are referred to as wide-sense sparse (or less sparse) multipath channels, as they can still be equalized using certain sparse estimation algorithms. In such channels, a greater number of resolvable paths exist, with the energy more diffusely distributed [55], [56].

For the two channel conditions, the simulated channels are randomly generated according to (3). The random interarrival time of consecutive paths, $\tau_{p+1} - \tau_p \forall p$, is exponentially distributed with a mean of 1.5 ms. The amplitudes $\{A_p\}_{p=1}^{N_{pa}}$ are independent and Gaussian distributed

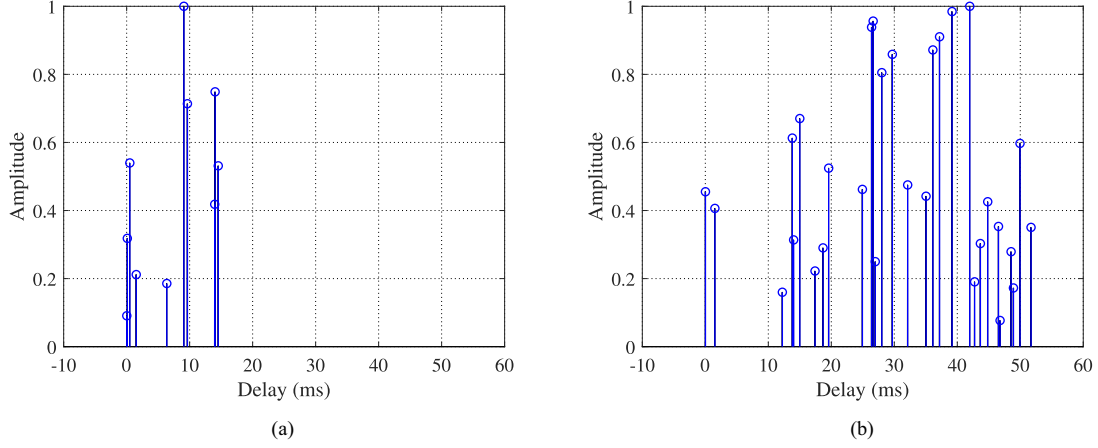


Fig. 2. Examples of CIRs in the simulation: (a) a very sparse multipath channel and (b) a wide-sense sparse multipath channel. The amplitudes are normalized.

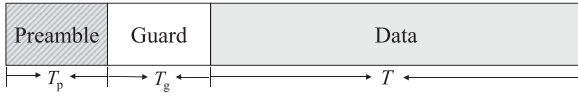


Fig. 3. Frame structure of the transmitted signal, including the preamble, guard interval, and data block.

TABLE II
PARAMETERS IN SIMULATION

Parameters		Values
Center frequency	f_c	10 kHz
Sampling frequency	f_s	48 kHz
Bandwidth of data	B	2 kHz
Symbol duration	T_{sym}	1 ms
Preamble interval	T_p	40 ms
Guard interval	T_g	100 ms

with average power decreasing exponentially with the delay of paths. Sparse channels have fewer paths ($N_{\text{pa}} = 10$) and a shorter average delay spread (15 ms), while wide-sense sparse channels have more paths ($N_{\text{pa}} = 30$) and a longer average delay spread (45 ms) [56], as shown in Fig. 2. Each channel condition is evaluated using 1000 Monte Carlo simulations.

In the simulation, the channel is assumed to be time-invariant within a single frame, meaning it remains constant during channel equalization and estimation. However, the channel varies from frame to frame.

The signaling frame in the simulation is shown in Fig. 3. An LFM signal is used as a preamble, with a start frequency of 12 kHz and a bandwidth of 4 kHz ahead of the transmitted signal. LFM is followed by a guard interval to mitigate ISI. LFM lasts for $T_p = 40$ ms, and the guard interval $T_g = 100$ ms. Each frame consists of 1000 binary phase-shift keying (BPSK) symbols, with each symbol lasting $T_{\text{sym}} = 1$ ms, resulting in a data duration $T = 1$ s. The frame duration $T_p + T_g + T = 1.14$ s. Rectangular pulse shaping is used in the simulation and experiment. The pulse duration equals the symbol duration, i.e., $\tau = 1$ ms, and the main lobe bandwidth of the passband data $B = 2/\tau = 2$ kHz. The detailed parameters are listed in Table II. The first 100 symbols of each frame are used as pilots. In Scheme 1 of our simulations, we use the cross-correlation of the LFM preamble for synchronization and channel estimation.

Fig. 4 illustrates the SER performance versus the step size D under channels in Fig. 2 without channel coding. Generally, Scheme 2 exhibits better performance than Scheme 1 with VTCMs in Fig. 4. Both figures demonstrate that the SERs vary with step size D in different channel conditions. A larger D does not necessarily lead to a lower SER. In either scheme and any channel condition, it is evident that an optimal D exists. Therefore, a suitable step size D can improve the performance, and it is reasonable to propose the RLVTM.

Fig. 5 shows the MSR and SER performance under the sparse channel in Fig. 2(a).

In Fig. 5(a), the MSRs versus E_b/N_0 are presented, where E_b is the energy of a single bit, N_0 is the power spectral density of the additive noise, and E_b/N_0 is equal to SNR in BPSK [58]. Scheme 2 still outperforms Scheme 1. We compare the proposed VTCM and RLVTM with the conventional VTRM. All the algorithms use the same pilots. The proposed VTCM and RLVTM exhibit higher MSRs than the conventional VTRM. Although the VTCM and RLVTM are designed with sparse estimation in the algorithm, both can deal with not only the sparse channel but also the wide-sense sparse channel in Fig. 2. The higher MSRs indicate that after equalization, the energy is more focused on the main path in the equalized channel. In Fig. 5(b), SER curves versus E_b/N_0 are provided. VTCM and RLVTM exhibit lower SERs than VTRM, which is consistent with the equalization performance measured by MSRs. We also include classical decision feedback equalizer (DFE) algorithms, such as LMS and RLS, as well as improved versions with sparse considerations, such as IPNLMS and PRLS, into the comparisons. The LMS algorithm uses a step size μ of 0.005, and the RLS algorithm applies a forgetting factor λ of 0.998 and δ of 0.5 [5]. For the improved algorithms, the IPNLMS algorithm is configured with a step size μ of 0.2, δ of 10^{-4} , and β of 0 [8], while the PRLS algorithm uses the same forgetting factor λ and δ as RLS, with a step size μ of 15, β of 0.5, and the regularized parameter $\varepsilon = 10^{-4}$ [46]. However, classical LMS exhibits the highest SER among all methods. Although the PRLS algorithm considers the sparsity of UWA channels, its performance is not as good as VTCM and RLVTM.

Among the equalization algorithms, the proposed RLVTM in Scheme 2 achieves the highest MSRs and correspondingly the lowest SER, owing to its optimization of the step size through interactions with the UWA environment through RL. It is closely followed by RLVTM in Scheme 1, which also performs efficiently. Scheme 1 of VTCM, while still effective, generally performs less favorably compared to Scheme 2. The SER of VTCM with $D = 3$ is initially slightly lower than that of $D = 6$, but as E_b/N_0 increases, it becomes slightly higher.

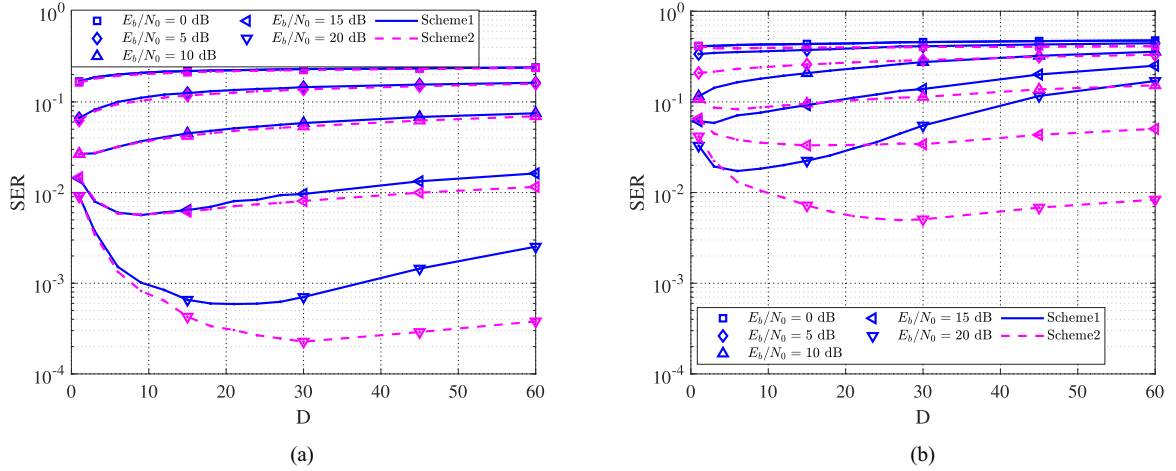


Fig. 4. SER performance of VTCM versus the step size D (a) under the sparse channel as in Fig. 2(a), and (b) under a wide-sense sparse channel, as in Fig. 2(b). Scheme 2 has a better performance than Scheme 1 in both channel conditions. The bigger step size D does not lead to the lower SER. A suitable D exists.

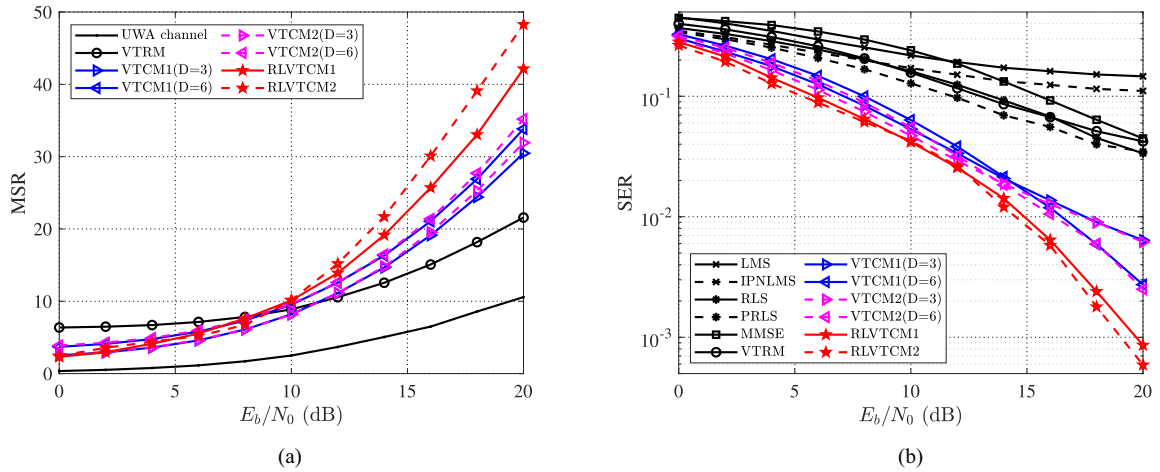


Fig. 5. (a) MSR performance and (b) SER performance versus E_b/N_0 under sparse channel in Fig. 2(a). Generally, the SER performance is consistent with the MSR performance: the higher the MSR is, the lower the SER is. Traditional equalization methods (MMSE, LMS, IPNLMS, RLS, and PRLS), both with and without sparse consideration are taken into the comparisons in SER performance. In both criteria, RLVTM in Scheme 2 shows significant improvement in equalization performance.

This indicates that keeping D constant does not necessarily result in a consistently lower SER across all E_b/N_0 values. A time-varying $D(n)$ is needed as in RLVTM.

Generally, the SER performance in Fig. 5(b) is consistent with the MSR performance in Fig. 5(a): the higher the MSR is, the lower the SER is. This consistency indicates that the proposed MSR is a good measure of the equalization performance.

Fig. 6 illustrates the MSR and SER performance of the proposed algorithms under the wide-sense sparse channel, as in Fig. 2(b). The main path of the channel in Fig. 2(b) is noticeably weaker than the side paths, which usually leads to lower MSRs and more severe ISI than those in the channel in Fig. 2(a).

Due to the denser multiple paths of the channel in Fig. 2(b) and longer multipath propagation, all the equalizers exhibit higher SERs than in Fig. 5. However, the proposed algorithms are still better than conventional equalization. This is owing to the embedded GOMP algorithm, which is a sparse estimation algorithm but also can work well in dense channels. In addition, the SERs with Scheme 1 in the proposed VTCM and RLVTM increase more than Scheme 2 due to

the challenges in channel estimation. However, the proposed VTCM and RLVTM in Scheme 2 still show the lowest SERs among the equalizers. The SER performance in Fig. 6(b) is still very consistent with the MSR performance in Fig. 6(a) under this channel condition.

VII. EXPERIMENTAL RESULTS AND ANALYSIS

The real-lake UWA communication experiment was conducted in 2022 near Zhanjiang, Guangdong, China. The data were also used in the International Underwater Robotics Competition, 2022. The experiment is a lake trial conducted in shallow water, with the transmitter and receiver deployed at the isothermal layer in the water (at depths of around 7–20 m). A single-element acoustical transducer and a single hydrophone were utilized during the experiment. Both the transmitter and receiver were mounted on floating platforms, allowing them to drift naturally. The water conditions were calm (Beaufort scale 0–2) with negligible surface wave activity and no significant currents. The measured Doppler shifts range from 0.1 to 10 Hz. Although the maximum Doppler shift reaches 10 Hz, most of the data exhibit Doppler shifts

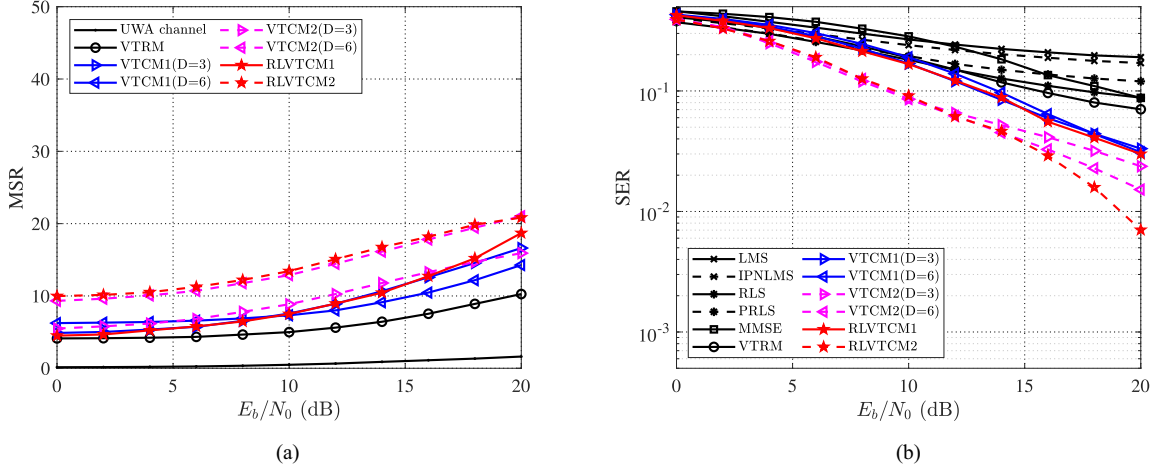


Fig. 6. (a) MSR performance and (b) SER performance versus E_b/N_0 under wide-sense sparse channel in Fig. 2(b). Scheme 2 still outperforms Scheme 1 in both VTCM and RLVTSM, demonstrating greater robustness and efficiency. RLVTSM (Scheme 2) achieves the lowest SER. The alignment of SER results with MSR performance further validates the effectiveness of the MSR criterion in assessing equalization performance.

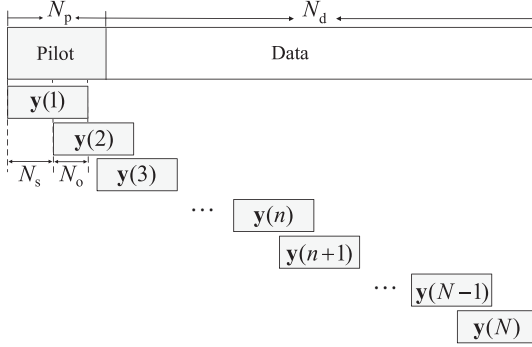


Fig. 7. Block diagram of the iterative receiver [43].

below 2 Hz. The transmitter–receiver distances range from hundreds of meters to several kilometers in the experiments. The received SNRs range from 10 to 38 dB, while the maximum delay spreads range from 5 to 30 ms.

The channel exhibits slight time variations within a frame due to the floating motion of the transmitter and receiver. To address these variations, an iterative receiver [43] is employed. As shown in Fig. 7, the received signal within a frame is divided into overlapping short blocks with length N_b and adjacent blocks overlap N_o symbols and $N_b = N_s + N_o$, where N_s is the head length of the block. The data are processed block by block iteratively: overlapping symbols N_o propagate refined channel estimates to the $n + 1$ th block for iterative equalization due to channel coherence.

Among the received data, we consider two typical underwater channel conditions. The received SNR was about 35.39 and 26.46 dB, respectively. To obtain the time-varying CIRs in Fig. 8, the OMP algorithm is used for channel estimation, along with an iterative receiver [43] to track the variation.

In Fig. 8(a), the channel is clearly sparse. The energy disperses in two most powerful paths, and it usually makes channel equalization difficult. The time delay spread in this channel is approximately 9 ms, indicating that the ISI spans over more than nine symbols. In Fig. 8(b), the channel has dense paths, which can be taken as a wide-sense sparse channel. The energy is scattered into many paths. The time spread delay

TABLE III
PARAMETERS IN EXPERIMENT

Parameters	Values
Center frequency	f_c 10 kHz
Sampling frequency	f_s 128 kHz
Bandwidth of data	B 2 kHz
Symbol duration	T_{sym} 1 ms
Preamble interval	T_p 40 ms
Guard interval	T_g 78.125 ms

in this channel is about 15 ms. Both channels are challenging for UWA communications.

In the experiment, the frames have the same structure as the simulations [57] but with different parameters. The sampling frequency is 128 kHz. The guard interval is 78.125 ms, and the data last for 1 s. Each frame contains 1000 BPSK symbols. The parameters are listed in Table III.

In the experimental results, white Gaussian noise with appropriate variance is randomly generated and added to the received signals to obtain signals corresponding to different E_b/N_0 values in each realization. For each E_b/N_0 , the SER is averaged over 1000 symbols and across 200 realizations.

For the traditional methods, the parameter settings of LMS and IPNLMS are identical to those used in the simulation. The forgetter parameter $\lambda = 0.97$ and $\delta = 100$ in both RLS and PRLS algorithms. Step size $\mu = 10$, weight $\beta = -0.5$, and the regularized parameter $\varepsilon = 10^{-4}$ in PRLS.

Fig. 9 illustrates the MSRs and SERs versus E_b/N_0 under the real-lake channel in Fig. 8(a). Generally, the results are consistent with simulations. In Fig. 9, the SERs of proposed VTCM and RLVTSM are lower than classical DFE algorithms. RLVTSM in Scheme 2 exhibits the lowest SER owing to the learning ability of the RL. RLVTSM in Scheme 1 has the second lowest SER.

Both RLVTSM and VTCM demonstrate a larger gap between SERs in Schemes 1 and 2 than in simulations. This is because the channel estimation performance in the real-lake experiment is usually worse than in the simulations, and Scheme 2 can avoid channel estimation.

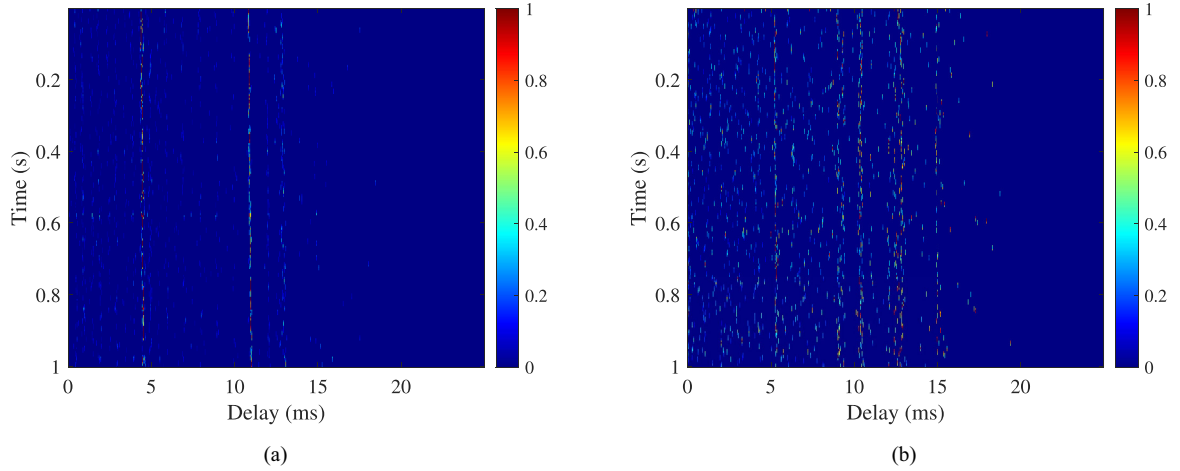


Fig. 8. Measured impulse responses of two typical UWA channels obtained through OMP estimation in the real-lake experiment: (a) very sparse multipath channel and (b) wide-sense sparse multipath channel. The amplitudes are normalized.

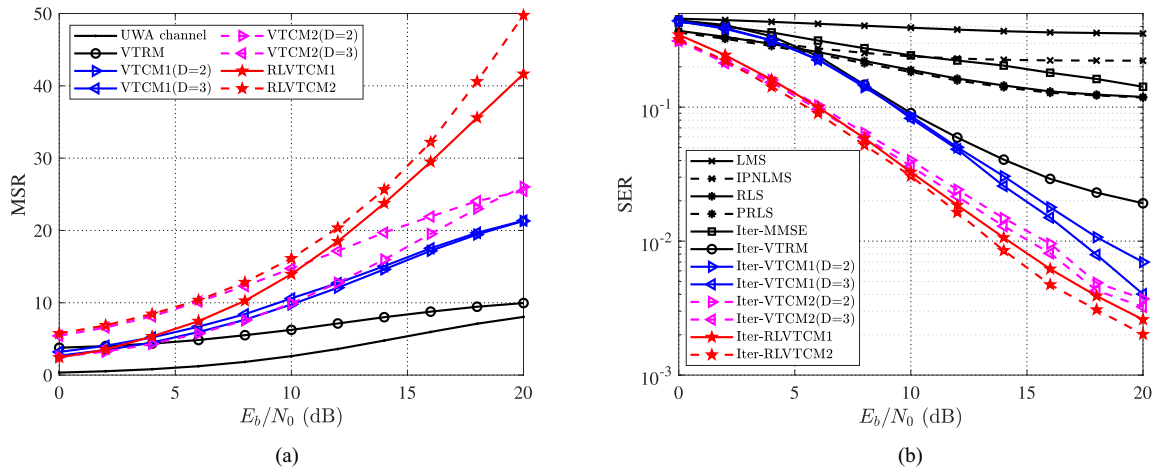


Fig. 9. (a) MSR performance and (b) the SER performance change with the E_b/N_0 under the very sparse channel as in Fig. 8(a). RLVTCM in Scheme 2 outperforms all other algorithms in different E_b/N_0 . The MSR and SER results are very consistent in this real-lake experiment.

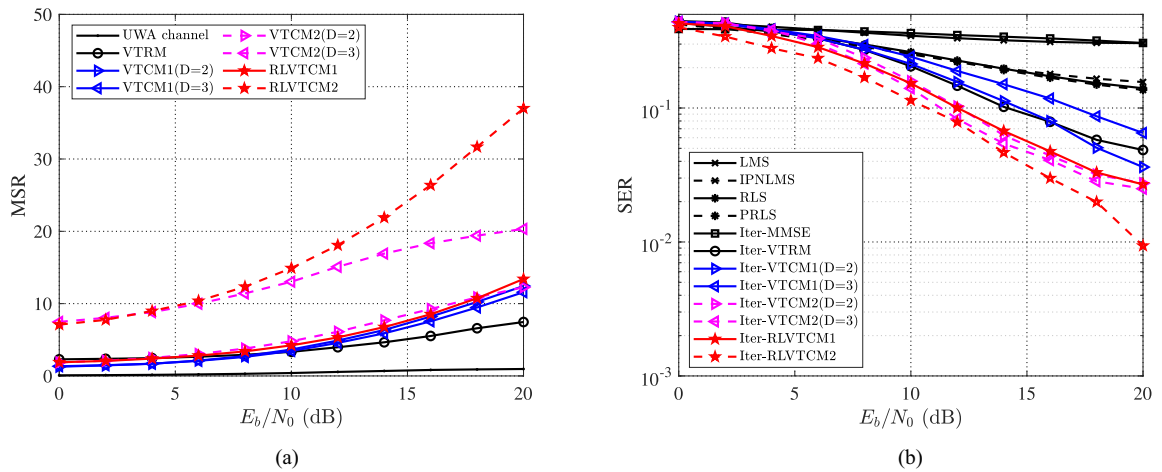


Fig. 10. (a) MSR and (b) SER performance versus E_b/N_0 under the channel, as in Fig. 8(b).

Conventional equalizers, such as LMS, RLS, and PRLS, exhibit degraded performance compared to simulations. It can be observed that the MSR trends in Fig. 9(b) align with the SER performance in Fig. 9(a) in these real-lake experiments.

Fig. 10 presents the MSRs and SERs versus E_b/N_0 under the wide-sense sparse channel in Fig. 8(b). Despite multipath effects becoming more severe in Fig. 8(b), the proposed RLVTM and VTCM achieve significantly lower SERs than classical DFE algorithms. RLVTM in Scheme 2 still exhibits the lowest SER. The MSR performance in Fig. 10(b) remains consistent with the SER performance in Fig. 10(a) in the real-lake wide-sense sparse channels.

VIII. CONCLUSION

In this work, we propose a VTCM approach in the temporal compression sense for UWA channel equalization. It offers a robust alternative to traditional methods, which usually rely on the performance of the channel estimation.

We also propose the MSR criterion in the temporal compression sense for UWA channel equalization. It effectively accounts for ISI using the PDP of the multipath channels.

Furthermore, RL is introduced into VTCM, thereby RLVTM is provided. It takes the proposed MSR as a reward to optimize step size and sparsity parameters in real time to adapt dynamically to changing UWA channel conditions.

Our simulations and experiments confirm that VTCM and RLVTM outperform traditional equalization algorithms, including VTRM, MMSE, LMS, IPNLMS, RLS, and PRLS. Among the proposed VTCM and RLVTM, the pilot-employed RLVTM shows significant improvement in channel equalization. It avoids channel estimation and achieves the highest MSR and lowest SER without channel coding.

For future perspectives, besides the iterative receiver, we will consider using RL to track the Doppler spread parameters in channel equalization, integrating coding schemes and further improve the RLVTM in Scheme 2. More real-sea experiments will be conducted with the multicarrier modulation systems.

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